quantasome contains about 230 chlorophyll molecules. A minimum number of chlorophyll and other accessory pigments act together in a photochemical reaction to release one oxygen or to reduce one molecule of CO_2. It constitutes a photosynthetic unit. (Figure 13.6) **Emerson** and **Arnold** (1932) based on flashing light experiment found 2500 chlorophyll molecules are required to fix one molecule of CO_2. However, the reduction or fixation of one CO_2 requires 10 quanta of light and so each unit would contain 1/10 of 2500 i.e. 250 molecules. Usually 200 to 300 chlorophyll molecules are considered as a physiological unit of photosynthesis. According to Emerson 8 quanta of light are required for the release of one oxygen molecule or reduction of one Carbon dioxide molecule. The quantum yield is 1/8 or 12 %.

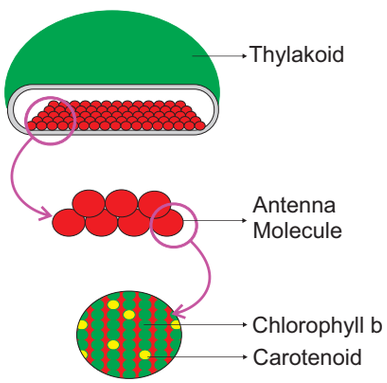


Figure 13.6: Quantasome

13.5 Absorption Spectrum and Action Spectrum

13.5.1 Absorption Spectrum

The term absorption refers to complete retention of light, without reflection or transmission. Pigments absorb different wavelengths of light. A curve obtained by plotting the amount of absorption of different wavelengths of light by a pigment is called its **absorption spectrum**.

- Chlorophyll 'a' and chlorophyll 'b' absorb quanta from blue and red region
- Maximum absorption peak for different forms of chlorophyll 'a' is 670 to 673, 680 to 683 and 695 to 705nm.
- Chlorophyll 'a' 680 (P680) and Chlorophyll 'a' 700 (P700) function as trap centre for PS II and PS I respectively.

13.5.2 Action Spectrum

The effectiveness of different wavelength of light on photosynthesis is measured by plotting against quantum yield. The curve showing the rate of photosynthesis at different wavelengths of light is called **action spectrum**. From the graph showing action spectrum, it can be concluded that maximum photosynthesis takes place in blue and red region of the spectrum. This wavelength of the spectrum is the absorption maxima for Chlorophyll (a) and Chlorophyll (b). The Action Spectrum is instrumental in the discovery of the existence of two photosystems in O_2 evolving photosynthesis (Figure 13.7).

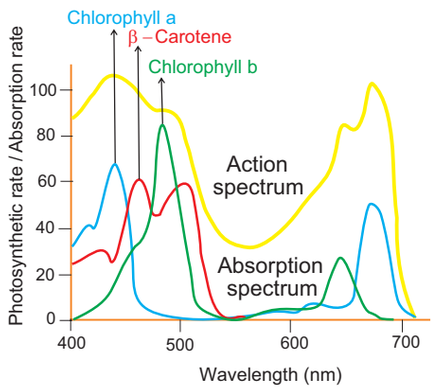


Figure 13.7: Absorption and action spectrum

13.6 Emerson's Experiments and Hill's Reaction

13.6.1 Red Drop or Emerson's First Effect

Emerson conducted experiment in *Chlorella* using only one wavelength of light (monochromatic light) at a time and he measured quantum yield. He plotted a graph of the quantum yield in terms of O_2 evolution at various wavelengths of light. His focus was to determine at which wavelength the photochemical yield of oxygen was maximum. He found that in the wavelength of 600 to 680 the yield was constant but suddenly dropped in the region above 680 nm (red region). The fall in the photosynthetic yield beyond red region of the spectrum is referred as **Red drop or Emerson's first effect.**

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